

Air Conditioning (A/C) Compressor



Material	
Item	Specification
Motorcraft® PAG Refrigerant Compressor Oil YN-12-D	WSH-M1C231-B

Removal and Installation

NOTICE: If installing a new Air Conditioning (A/C) compressor due to an internal failure of the old unit, the following procedures must be carried out to remove contamination from the A/C system. Failure to remove contamination from the A/C system, if present, will result in poor A/C performance and/or damage to the new A/C compressor and other components.

- Carry out flushing of the A/C system prior to installing a new A/C compressor. For additional information, refer to Section 412-00.
- Install a new Thermostatic Expansion Valve (TXV) , as directed by the A/C flushing procedure.
- Install a new receiver/drier as directed by the A/C flushing procedure.

NOTE: Installation of a new receiver/drier is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier. Damage to the receiver/drier includes leaks in the receiver/drier, physical damage to the receiver/drier shell or desiccant, or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier should not be replaced.

All vehicles

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. If flushing of the A/C system has not been carried out, recover the refrigerant. For additional information, refer to Section 412-00.

3.5L GTDI vehicles

3. Remove the Air Cleaner (ACL) outlet pipe, RH ACL pipe to-turbocharger pipe, RH turbocharger bypass hose and the RH turbocharger-to-Charge Air Cooler (CAC) pipe. For additional information, refer to Intake Air System Components — Exploded View, 3.5L Gasoline Turbocharged Direct Injection (GTDI) in Section 303-12.

All vehicles

4. Disconnect the clutch field coil electrical connector.
5. Detach the wire harness retainer from the A/C compressor and position the wire harness aside.
6. Remove the 2 A/C compressor fitting nuts and disconnect the fittings.
 - Discard the O-ring seals and gasket seals.
 - To install, tighten to 15 Nm (133 lb-in).

7. If equipped, remove the front skid-plate.
8. Remove the drive belt from the A/C compressor pulley. For additional information, refer to Section 303-05.
9. Remove the 3 A/C compressor bolts.
 - To install, tighten to 25 Nm (18 lb-ft).
10. Remove the A/C compressor.
11. To install, reverse the removal procedure.
 - If a new A/C compressor is to be installed, the clutch assembly may need to be transferred from the old unit to the new unit. For additional information, refer to Clutch and Clutch Field Coil in this section.
 - Install new gasket seals and O-ring seals.
 - Add the correct amount of clean PAG oil to the refrigerant system. For additional information, refer to the Refrigerant Oil Adding procedure in Section 412-00.
12. Evacuate, leak test and charge the refrigerant system. For additional information, refer to Section 412-00.